Example 2.14

Show that the points 1, $\frac{-1}{2}+i\frac{\sqrt{3}}{2}$, and $\frac{-1}{2}-i\frac{\sqrt{3}}{2}$ are the vertices of an equilateral triangle.

Solution

It is enough to prove that the sides of the triangle are equal.

Let $z_1=1,\ z_2=\frac{-1}{2}+i\frac{\sqrt{3}}{2}$, and $z_3=\frac{-1}{2}-i\frac{\sqrt{3}}{2}$.

The length of the sides of the triangles are

$$|z_1-z_2| = \left|1-\left(\frac{-1}{2}+i\frac{\sqrt{3}}{2}\right)\right| = \left|\frac{3}{2}-\frac{\sqrt{3}}{2}i\right| = \sqrt{\frac{9}{4}+\frac{3}{4}} = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

$$|z_2-z_3| = \left|\left(\frac{-1}{2}+i\frac{\sqrt{3}}{2}\right)-\left(\frac{-1}{2}-i\frac{\sqrt{3}}{2}\right)\right| = \sqrt{\left(\sqrt{3}\right)^2} = \sqrt{3}$$

$$|z_3-z_1| = \left|\left(\frac{-1}{2}+i\frac{\sqrt{3}}{2}\right)-1\right| = \left|\frac{-3}{2}-\frac{\sqrt{3}}{2}i\right| = \sqrt{\frac{9}{4}+\frac{3}{4}} = \sqrt{3}$$

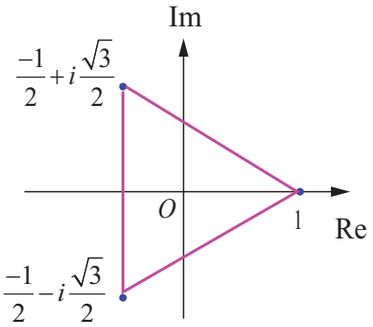


Fig. 2.21

Since the sides are equal, the given points form an equilateral triangle. ■

Example 2.15

Let z_1, z_2, and z_3 be complex numbers such that $|z_1|=|z_2|=|z_3|=r>0$ and $z_1+z_2+z_3\neq 0$.

Prove that $\left|\frac{z_1z_2+z_2z_3+z_3z_1}{z_1+z_2+z_3}\right|=r$.

Solution

Given that $|z_1| = |z_2|=|z_3|=r \Rightarrow z_1\bar{z}_1=z_2\bar{z}_2=z_3\bar{z}_3=r^2$

$$\Rightarrow z_1 = \frac{r^2}{\bar{z}_1},\quad z_2=\frac{r^2}{\bar{z}_2},\quad z_3=\frac{r^2}{\bar{z}_3}$$

Therefore $z_1+z_2+z_3 = \frac{r^2}{\bar{z}_1}+\frac{r^2}{\bar{z}_2}+\frac{r^2}{\bar{z}_3}$

$$= r^2\left(\frac{\bar{z}_2\bar{z}_3+\bar{z}_1\bar{z}_3+\bar{z}_1\bar{z}_2}{\bar{z}_1\bar{z}_2\bar{z}_3}\right)$$

$$|z_1+z_2+z_3| = |r^2|\left|\frac{\overline{z_2z_3+z_1z_3+z_1z_2}}{\overline{z_1z_2z_3}}\right| \qquad (\because \bar{z}_1+\bar{z}_2=\overline{z_1+z_2})$$

$$= r^2\frac{|z_2z_3+z_1z_3+z_1z_2|}{|z_1||z_2||z_3|} \qquad (\because\ |z|=|\bar{z}| \text{ and } |z_1z_2z_3|=|z_1||z_2||z_3|)$$

$$|z_1+z_2+z_3| = r^2\frac{|z_2z_3+z_1z_3+z_1z_2|}{r^3} = \frac{|z_2z_3+z_1z_3+z_1z_2|}{r}$$

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